

- ii. OR both the given message and pending acknowledgement are of **Unsecured Session Type**.
2. If there is a matching pending acknowledgement, the **A Flag** SHALL be set on the outbound message so it will serve as a piggybacked acknowledgement.
 - a. For such a piggybacked acknowledgement, the *Acknowledgment Message Counter* field SHALL be set to the message counter of the received message for which an acknowledgement was pending.
 - b. If the message being prepared is not a **standalone acknowledgement**, remove the matching entry from the *acknowledgement table*.
 - c. If the message being prepared is a **standalone acknowledgement**, set the **StandaloneAckSent** field of the matching entry in the *acknowledgement table* to true.

Message retransmission processing

1. If the outbound message is marked to be delivered reliably over a UDP transport, the **R Flag** SHALL be set on the given message to request an acknowledgement from the peer upon receipt.
 - a. Any message flagged for reliable delivery (**R Flag** set) SHALL be stored in the *retransmission table* to track the message until it has been successfully acknowledged by the peer.
2. Perform **Section 4.7.1, “Message Transmission”** processing step on the message to send the message to the peer:
 - a. The same Session ID, Destination Node ID, Security Flags, and transport as were used for the initial message transmission SHALL be used.
3. If the transport interface returns an error on the send attempt, the error is assessed to determine whether the message can be retried.
 - a. If the error is fatal, the application is notified and the message removed from the *retransmission table*.
 - b. If there is no error, or a non-fatal error such as no memory, the message is resent
 - i. Update the *retransmission table* to reflect the send count.
 - ii. Start a retransmission timer to track the maximum time to wait before attempting another retransmission.
 - iii. For each retry, the *retransmission table* is updated to track the number of retries until the maximum number is attempted, at which point the message is evicted from the *retransmission table*.

Send flow state diagram

The MRP send flow described above is depicted in the control flow diagram **Figure 12, “MRP send flow”**.